```
  for i in {0, .., maxIter-1} do
                                      #6
    newCentroids ← mapReduce(centroids, mappers, reducers)
                   #7
    if centroids == newCentroids then
                   #8
      break
    else
      centroids ←
newCentroids                                                    #9
  return combine(mappers,
centroids)                                              #10
```

#1 Method `MRkmeans` takes a list of points, the number of desired chunks in which the dataset should be sharded, the number of desired clusters, and two thresholds, `T1` and `T2`, to be used for canopy clustering initialization step.

#2 Break down the dataset into `numShards` random shards. This is usually done automatically by the master node of map reduce, but in this case, we will use a customized version.

#3 Initialize the centroids by using (a distributed version of) canopy clustering.

#4 Initialize `numShards` mappers, each will run the `classifyPoints` method and will hold a copy of one of the dataset's shards. The mappers will be alive (and hold the same copy of the input) for the whole lifetime of this method.

#5 Initialize `numClusters` reducers, one per centroid. The reducers will run the `centerOfMass` method (computing the center of mass of a set of points) and will be alive for the whole method's lifetime, but they won't hold any data.

#6 Repeat the main cycle at most `maxIter` times.